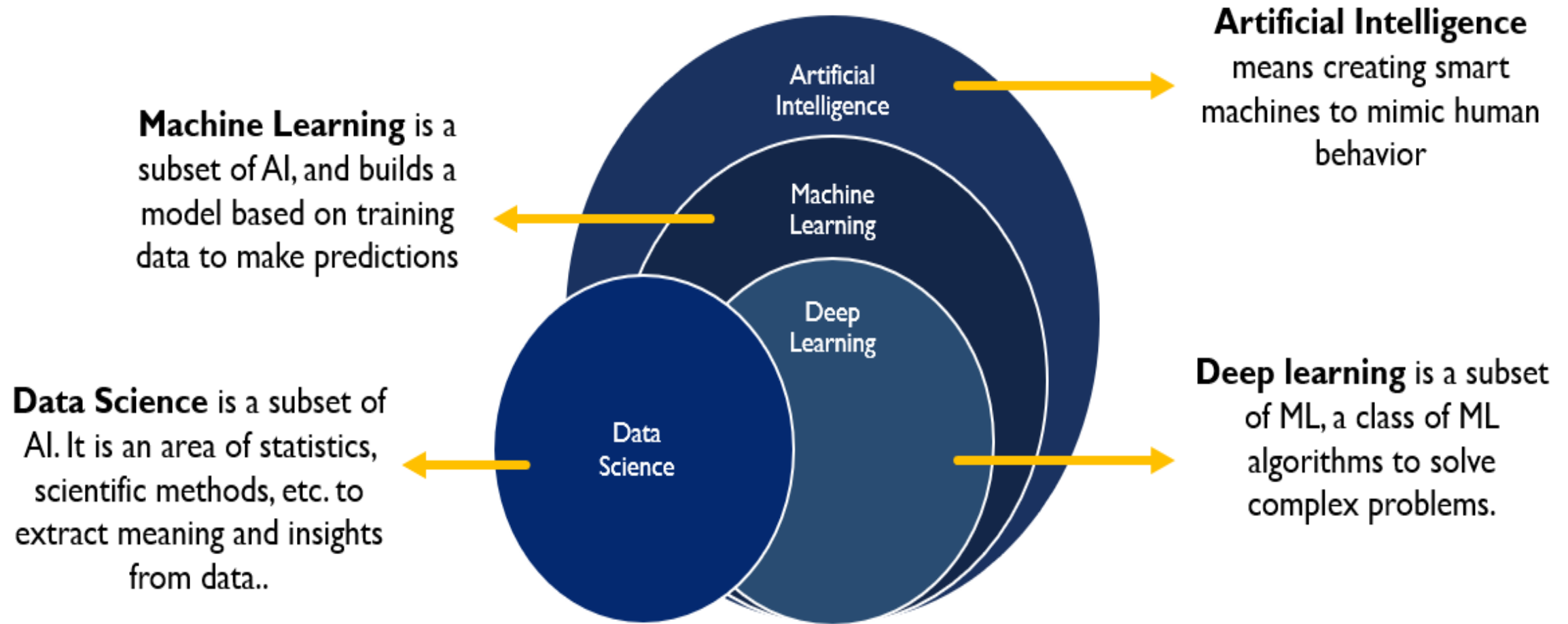




1.1 Intro to Machine Learning

❑ AI vs ML vs DL vs DS



1. Artificial Intelligence (AI)

Examples of AI:

- Chess-playing computer
- Siri / Google Assistant
- Self-driving car
- Chatbots
- Recommendation systems

2. Machine Learning (ML)

Examples of ML:

- Predicting house prices
- Email spam detection
- Fraud detection
- Weather forecasting

3. Deep Learning (DL)

Examples of DL:

- Face recognition in Instagram
- Self-driving cars detecting lanes
- ChatGPT (built on deep learning)
- Voice assistants understanding speech

4. Data Science (DS)

Examples of DS:

- Sales forecasting
- Customer behavior analysis
- Creating dashboards in Power BI / Tableau
- Identifying business trends

❑ Introduction to Machine Learning (ML)

- ✓ Machine Learning is a field of Artificial Intelligence where **computers learn from data** instead of being explicitly programmed.

➤ Traditional Programming

You give: Rules + Data → Output

➤ Machine Learning

You give: Data + Output → Model learns the Rules automatically

Ques. Why do we need Machine Learning?

Because many problems do not have simple rules.

Examples:

- How to recognize a face? (Millions of patterns)
- How to detect fraud in banking?
- How to translate English to Hindi?

Humans can do these easily by experience → ML gives that "experience" to computers.

Ques. How ML Works ?

Collect Data → Clean Data → Split (Train/Test)

Train a model → Evaluate → Improve → Deploy

Example: Predict house price

Data contains:

- Size of house
- Location
- Number of rooms
- Price

The ML model learns the relationship:

Bigger house = higher price

Prime location = higher price

Later, when you give a new house, the model predicts its price.

❏ Types of Machine Learning

➤ Supervised Learning

Model learns from **labeled data** (input, output given).

Examples:

- Predict house price
- Email spam detection
- Predict next day rainfall
- Classify handwritten digits (MNIST)

Algorithms: Linear Regression, Logistic Regression, Decision Trees, SVM, Neural Networks.

➤ Unsupervised Learning

Model finds patterns when **output is not given**.

Examples:

- Grouping customers with similar habits
- Finding clusters in data (K-Means)
- Reducing features (PCA)

Algorithms: K-Means, Hierarchical Clustering, PCA

➤ Reinforcement Learning

Model learns by **reward and punishment**.

Examples:

- Self-driving cars
- Playing chess/Go
- Robot navigation
- Game AI

❑ Real-Life Examples of ML

1. YouTube Recommendations

Learns what videos you like → suggests similar videos.

2. Google Maps ETA Prediction

Predicts travel time using past traffic data.

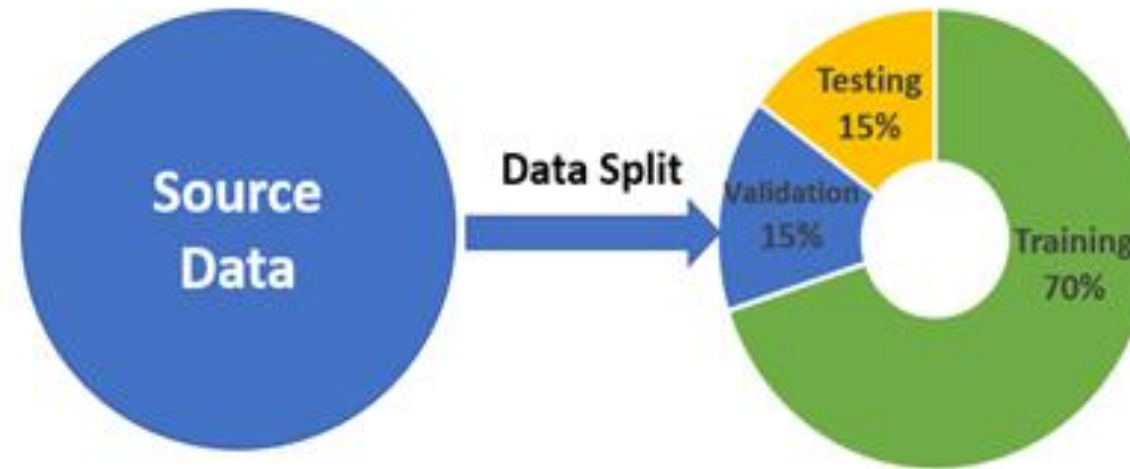
3. Face Unlock in Mobile

Learns your face features → matches in real-time.

4. Medical Diagnosis

Predicts diseases from X-ray/MRI images.

❑ Train, Test & Validation



How to split a dataset

TRAINING SET

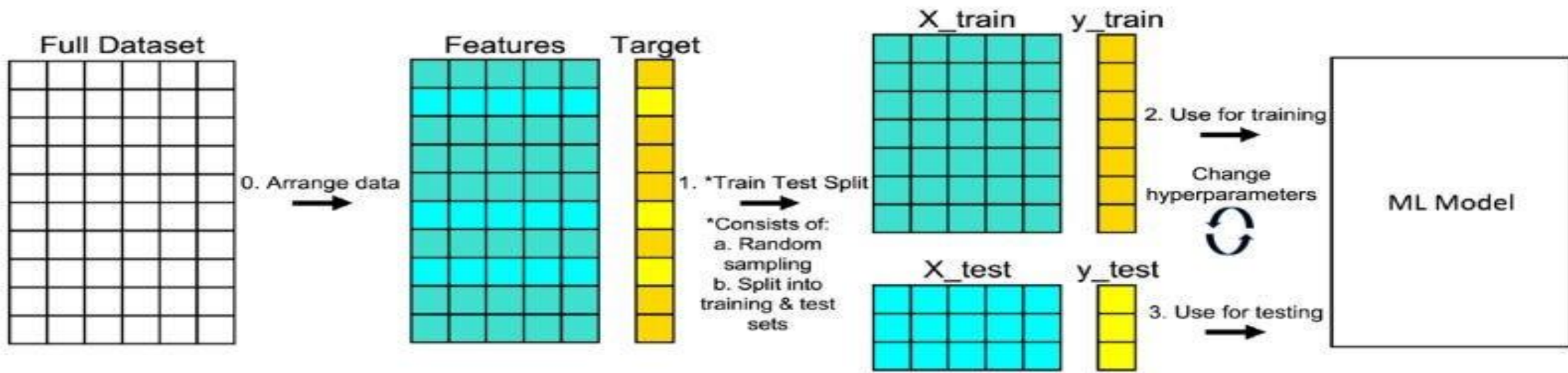
The subset of data used to train a machine learning model

TEST SET

The subset of data used to evaluate the performance of a trained machine learning model on unseen examples, simulating real-world data

VALIDATION SET

The intermediary subset of data used during the model development process to fine-tune hyperparameters



1. Training Set

The **training set** is the data used by the model to *learn patterns, weights, and relationships*.

“Training data is used to learn the model parameters.”

Example:

You want to predict house prices.

Training data: 70% of houses with price information.

The model learns:

- Bigger house → higher price
- Prime location → costly

2.Validation Set

Validation data is used to **tune the model** — choose the best hyperparameters.
It is **not used for learning**, only for checking performance during training.

Why we use validation set? for

- Selecting best learning rate
- Choosing number of layers
- Choosing degree of polynomial
- Early stopping
- Choosing regularization (λ)

You try:

- LR = 0.01 → accuracy = 82%
- LR = 0.1 → accuracy = 89%
- LR = 0.2 → accuracy = 76%

Example:

Out of 100 houses:

- 70 for training
- 15 for validation

LR → Learning rate

Best → LR = 0.1 found
from validation set.

“Validation set is used for hyperparameter tuning and model selection.”

3.Test Set

The **test set** is used *only once at the end* to measure **final accuracy** of the model on completely **unseen data**.

Example:

Remaining 15 houses are used as test set.
If the model predicts test houses correctly → model will work in real life.

“Test set estimates how well the final model generalizes to new data.”

✓ Simple Analogy

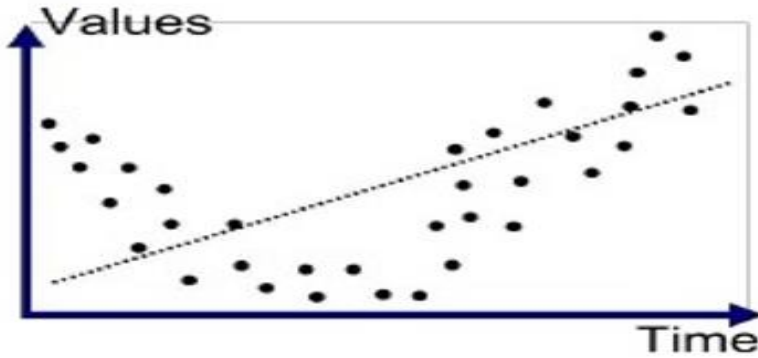
ML Concept	Real Life Meaning
Training Set	Chapters you study & practice from.
Validation Set	Mock tests you take to check performance & adjust strategy
Test Set	Final exam — questions you’ve never seen before.

Why do we need Validation & Test separately?

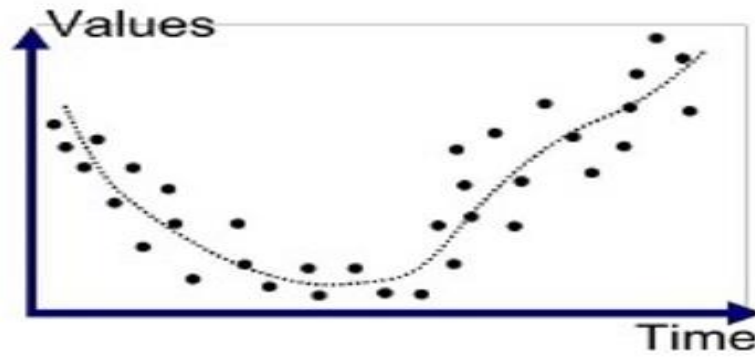
Because if you tune model using validation data, it gets indirectly optimized for it.
So we keep a **fresh, untouched test set** for final accuracy.

Train → Learn
Validation → Tune
Test → Evaluate Final
Performance on new data

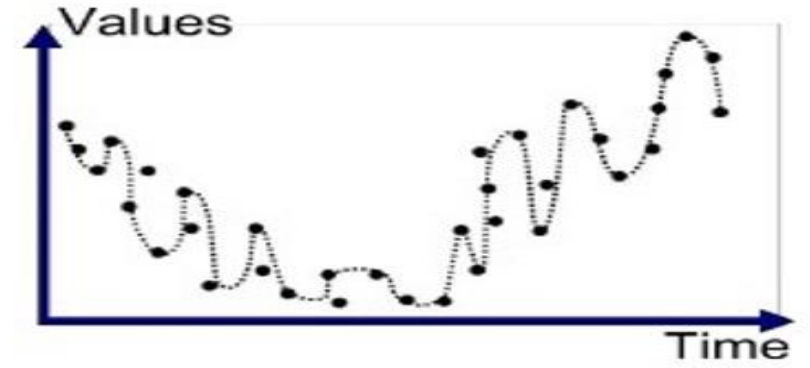
❑ Overfitting, underfitting & bias variance



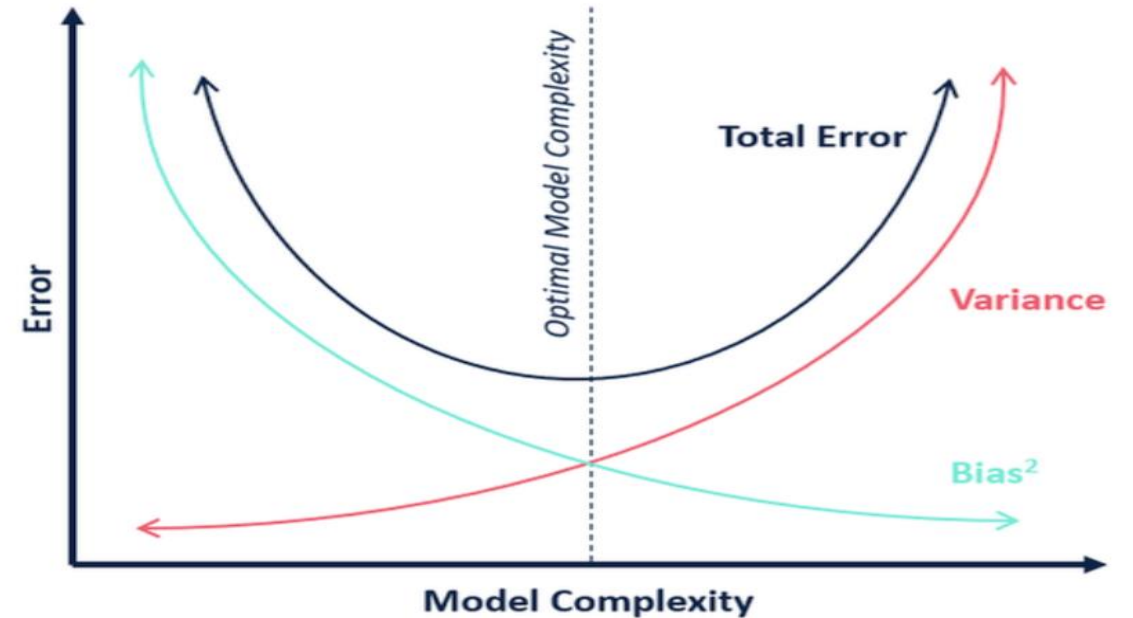
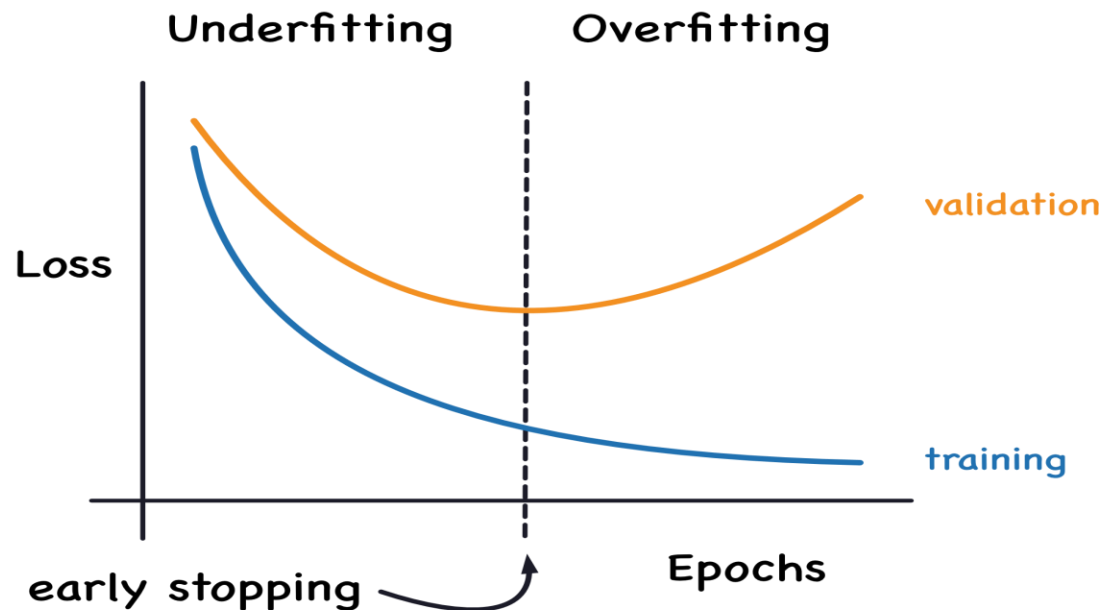
Underfitted



Good Fit/Robust



Overfitted



1. Underfitting

A model **too simple** to learn the underlying pattern.
It neither performs well on training data nor test data.

Easy Example

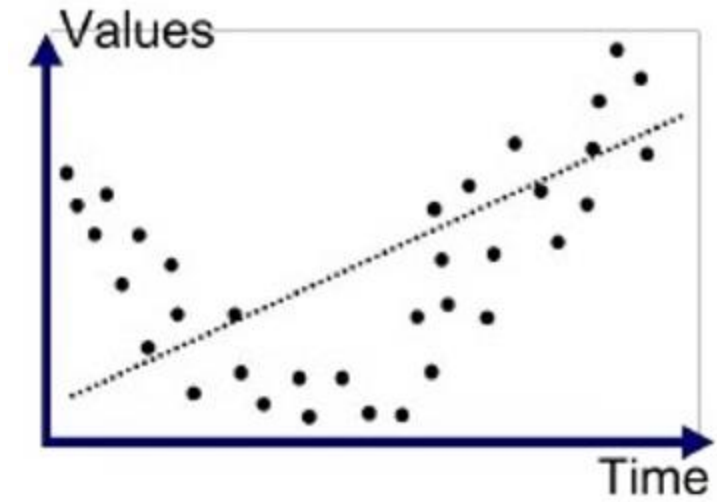
Data is curved (quadratic), but you fit a straight line.
The model cannot capture the trend → large errors.

Symptoms

- Low training accuracy
- Low test accuracy
- High bias (model is too simple)

Why it happens?

- Model too simple (Linear instead of polynomial)
- Not enough training
- Missing important features



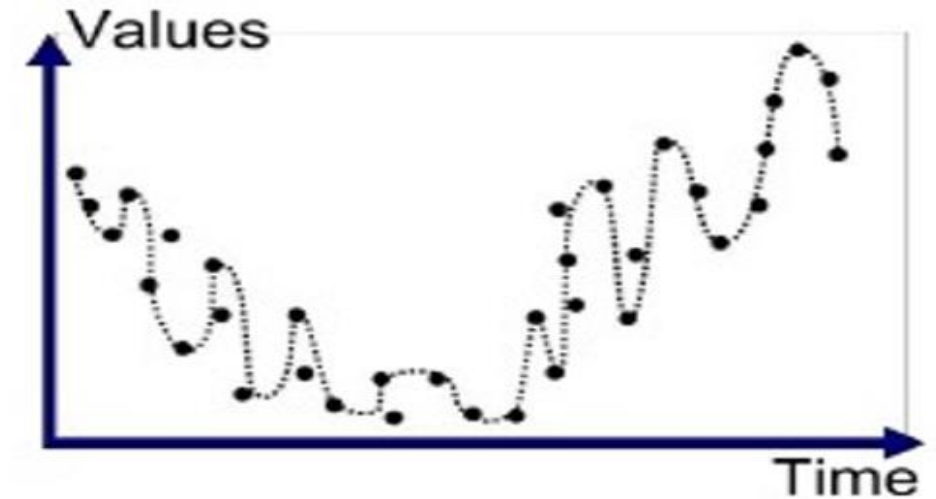
Underfitted

2. Overfitting

A model that **learns the training data too well**, even noise, but fails on new data.

Easy Example

Imagine drawing a line that passes exactly through every training point—even the noisy ones. It looks perfect on training data but bad on unseen data.



Overfitted

Symptoms

- High training accuracy
- Low test accuracy
- High variance (model too complex)

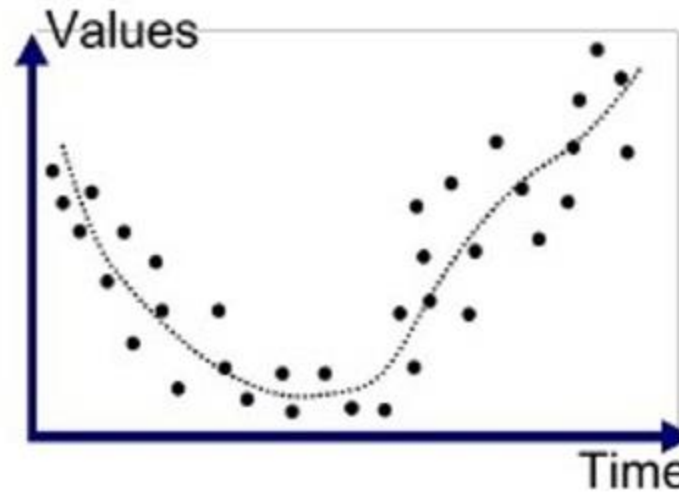
Why it happens?

- Too many parameters
- Model is overly flexible
- Small training dataset
- No regularization

3. Good Fit

A balanced model that learns the general pattern without memorizing.

- High training accuracy
- High test accuracy
- Balanced bias & variance



Good Fit/Robust

❑ Bias–Variance Tradeoff

✓ Bias

Error due to **simplifying the model too much**.

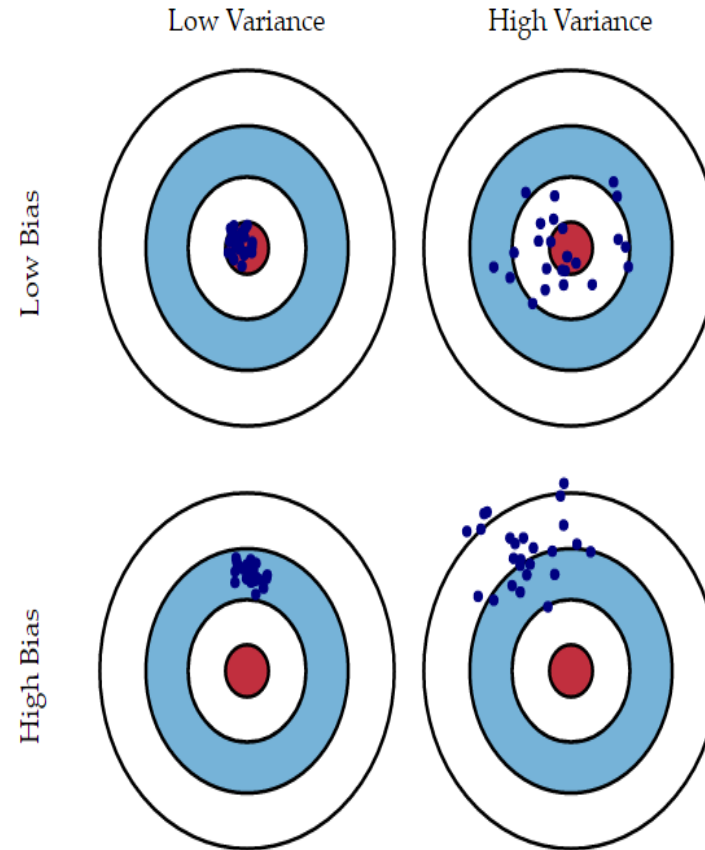
High bias $\rightarrow$ underfitting.

Examples:

- Linear model used for non-linear data
- Model ignores important features

Effect:

Training error $\uparrow$, Test error $\uparrow$



✓ Variance

Error due to **too much sensitivity to training data**.

High variance $\rightarrow$ overfitting.

Examples:

- Very deep neural network trained on small dataset
- High-degree polynomial model

Effect:

Training error $\downarrow$, Test error $\uparrow$

Bias–Variance Tradeoff (Easy Explanation)

Model Simplicity	Bias	Variance	Result
Too simple	High	Low	Underfitting
Just right	Medium	Medium	Good fit
Too complex	Low	High	Overfitting

“Bias is error from a simple model; variance is error from a complex model. We aim for a balance between both.”

➤ Real-Life Analogy

Underfitting

Reading only 2 pages before exam → you don't know enough → fail.

Overfitting

You memorize previous year's paper word-for-word → if questions change → fail.

Good Fit

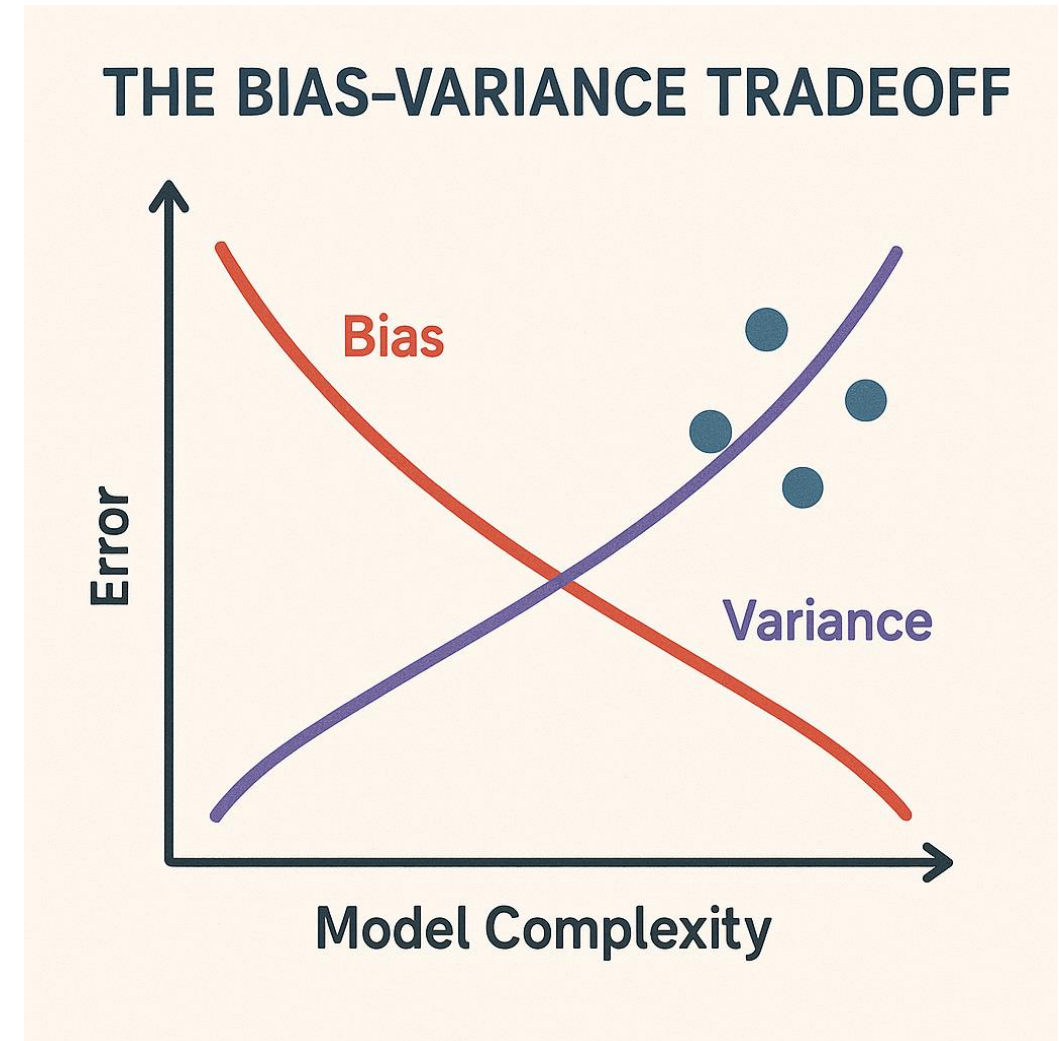
You understand concepts + practice some examples → perform well in exam.

➤ How to reduce Underfitting ?

- Increase model complexity
- Add more features
- Reduce regularization
- Train longer

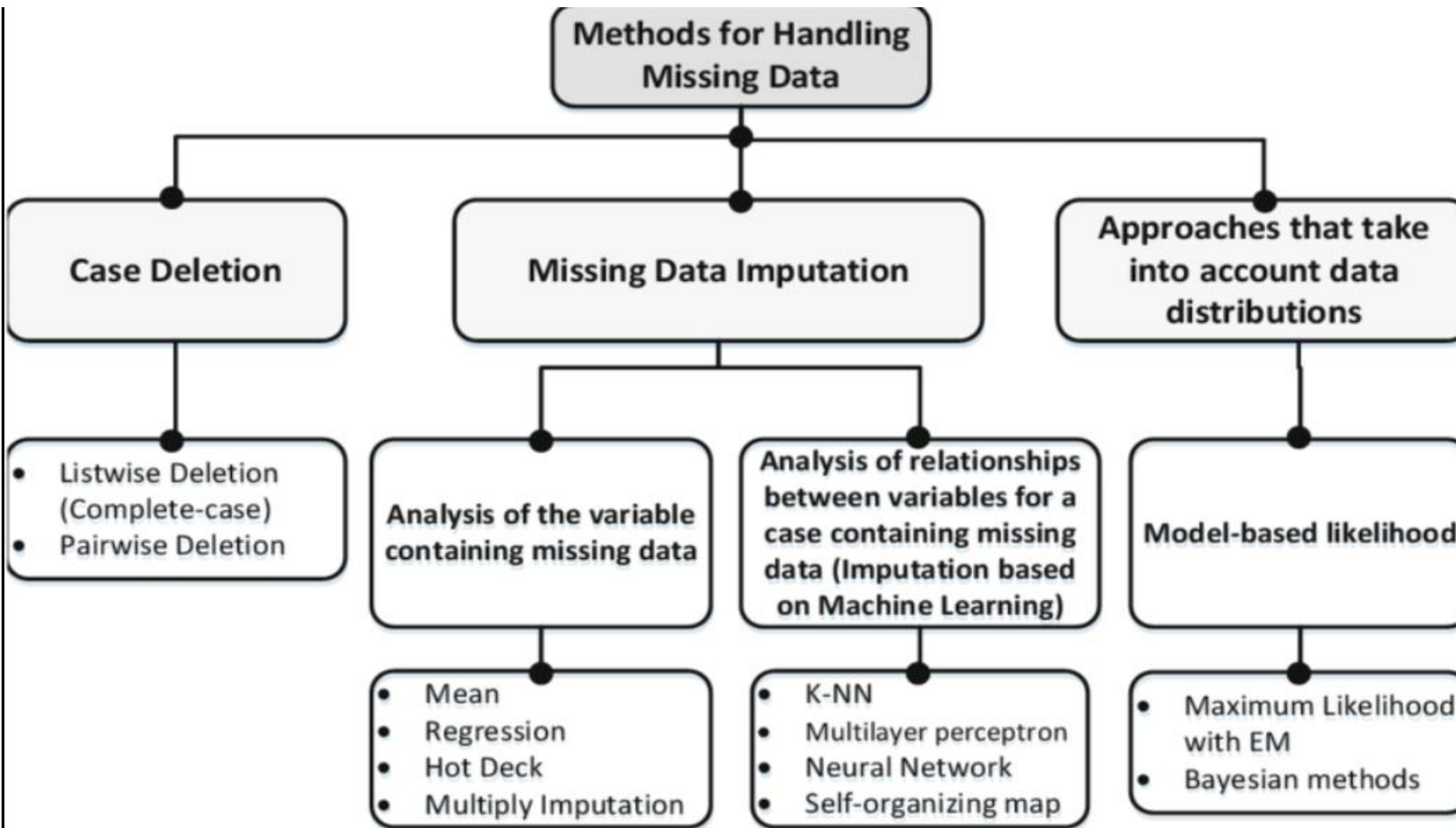
➤ How to reduce Overfitting ?

- Add more data
- Use regularization (L1/L2)
- Dropout (for neural networks)
- Simplify the model
- Early stopping



❑ Handling missing data

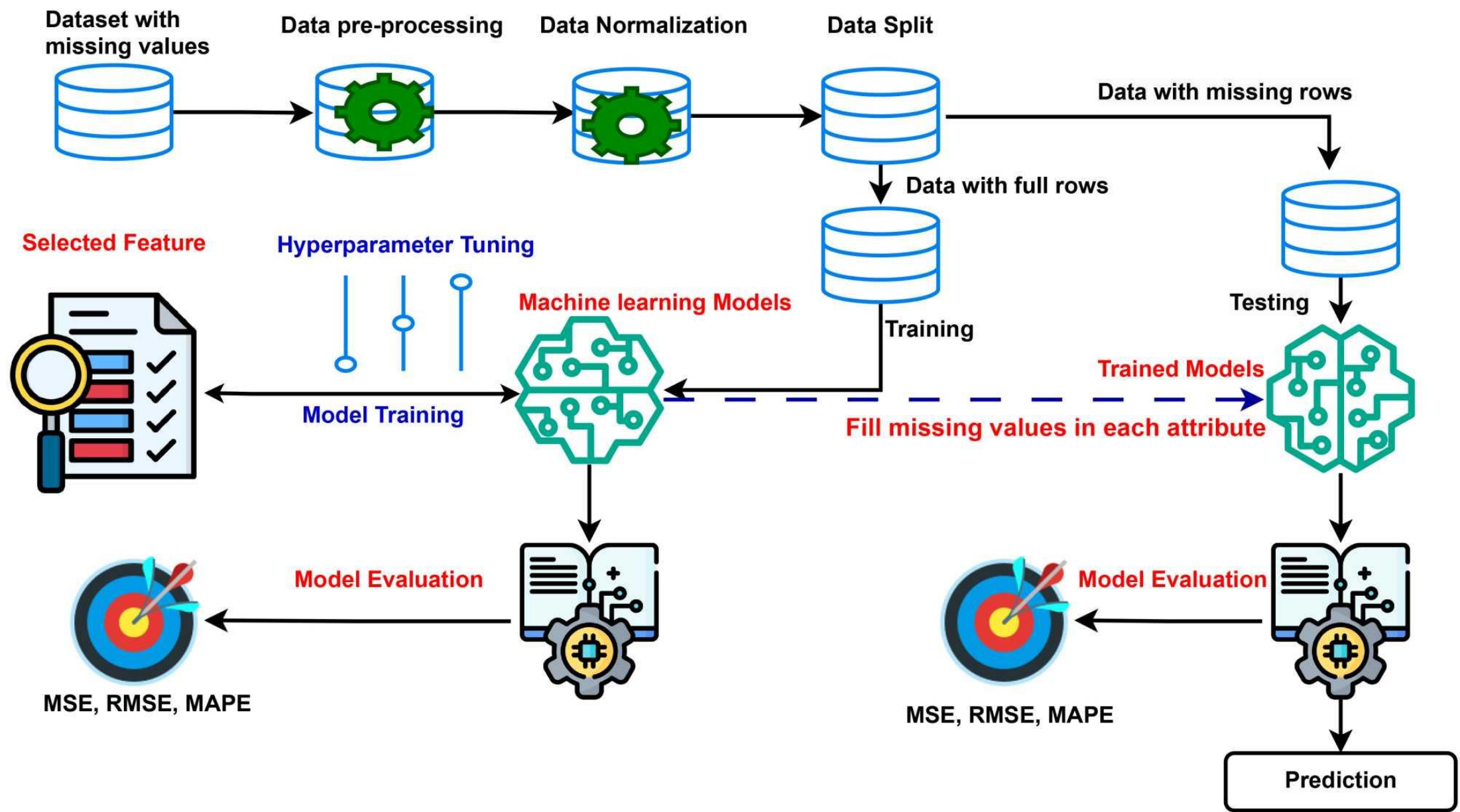
Missing data means some values in the dataset are **blank, null, NaN, or unavailable**.



Example from a weather dataset:

- Temperature = 30°C
- Humidity = *missing*
- Rain Tomorrow = Yes

❑ Handling missing data



➤ Why Missing Data is a Problem?

Because ML models cannot handle NaN values.

Missing values can cause:

- Wrong training
- Incorrect patterns
- Biased models
- Errors during model fitting

✓ Types of Missing Data

Type	Meaning	Example
MCAR (Missing Completely at Random)	No pattern	Sensor fails randomly
MAR (Missing At Random)	Depends on some variable	High earners skip salary column
MNAR (Missing Not At Random)	Missing due to its own value	People with high weight may skip weight entry

➤ Techniques to Handle Missing Data

1.Delete Rows (Listwise Deletion)

Remove rows where values are missing.

When to use

- Missing values < 5%
- Dataset is large
- Missing completely random (MCAR)

When NOT to use

- Small dataset
- Too many missing values
- Important information may be lost

2.Delete the Column

Remove entire feature with too many missing values.

Use when:

- Column has > 60–70% missing values
- Feature is not important

3.Imputation (Fill Missing Values)

(A) Mean Imputation

Replace missing values with **average**.

Use for:

- Numeric data, no outliers
- Example: age, height

(B) Median Imputation

Replace missing with **median** (middle value).

Use for:

- Numeric data with outliers
- Example: income, house price

(C) Mode Imputation

Replace missing values with **most frequent value**.

Use for:

- Categorical values
- Example: Gender, City

4. Advanced Imputation Methods

(A) KNN Imputation

Replace missing value using **nearest similar rows**.

Best when:

- Data has patterns
- Both numeric + categorical
- Example: medical data

(B) Regression Imputation

Predict missing values using another ML model.

Example: Predict missing "salary" using age, education, experience.

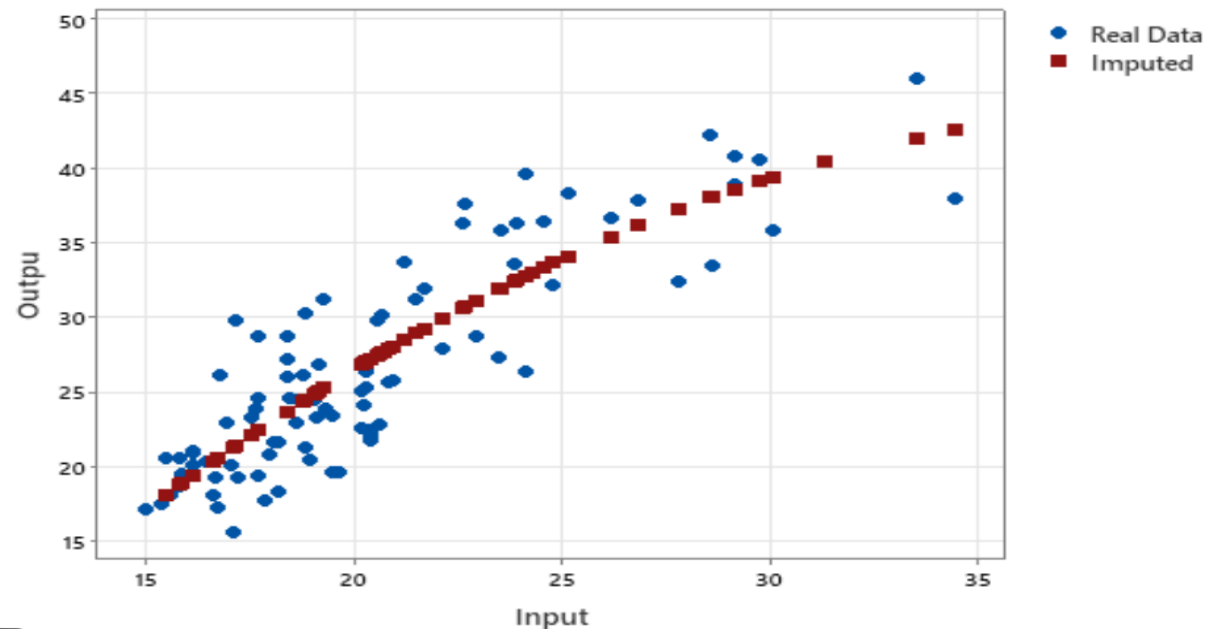
(C) Multiple Imputation

Creates multiple versions of data → averages results.

Example: Used in research/medical fields.

➤ Handling Missing Data in Pandas

```
df.dropna()                # delete rows
df.dropna(axis=1)          # delete columns
df.fillna(df.mean())       # mean imputation
df.fillna(df.median())     # median imputation
df['city'].fillna(df['city'].mode()[0]) # mode
```



Numerical Variables

- Mean/ Median Imputation
- Arbitrary Value Imputation
- End of tail Imputation
- Mode Imputation

Categorical Variable

- Frequent category Imputation
- Adding a "Missing" category

Both

- Complete Case Analysis
- Adding a "Missing" Indicator
- Random Sample Imputation

➤ Interview-Ready Quick Answers

Q1: Why is handling missing data important?

Because ML models cannot train on NaN values and missing data can introduce bias.

Q2: When do you use mean vs median?

- **Mean:** For normal data without outliers
- **Median:** When outliers exist

Q3: When do you delete rows?

When missing data $< 5\%$ and dataset is large.

Q4: How do you handle missing categorical data?

By mode imputation or creating a separate category like “Unknown”.